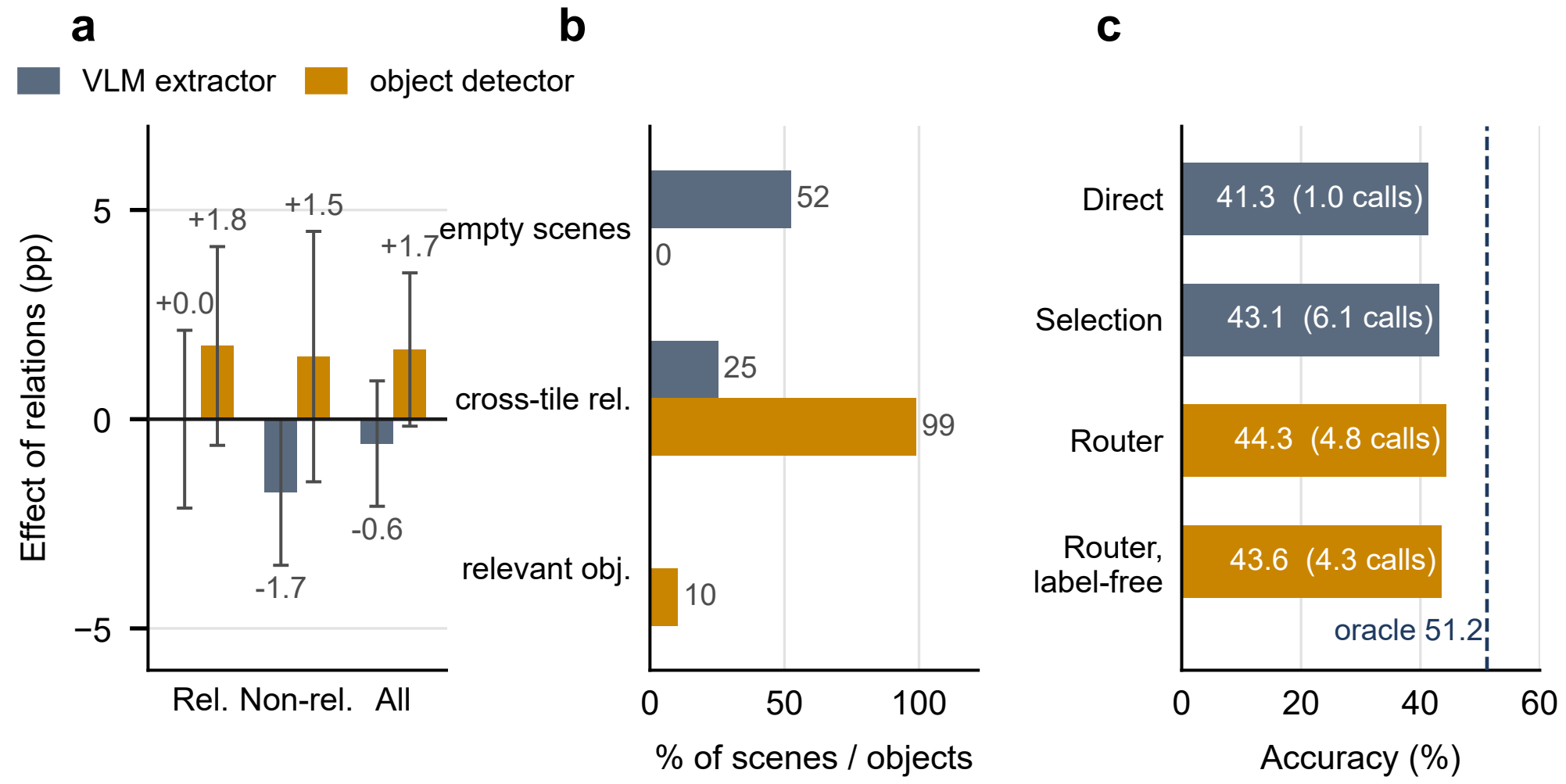
a
b
c
VLM extractor
object detector
Effect of relations (pp)
5
0
−5
+0.0
+1.8
+1.5
-1.7
-0.6
+1.7
Rel.
Non-rel.
All
empty scenes
cross-tile rel.
relevant obj.
52
0
25
99
10
0
50
100
% of scenes / objects
Direct
Selection
Router
Router, label-free
41.3 (1.0 calls)
43.1 (6.1 calls)
44.3 (4.8 calls)
43.6 (4.3 calls)
oracle 51.2
0
20
40
60
Accuracy (%)